
HCI Report

Human Computer Interaction (4IG05)

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1 Introduction

This report details the work conducted as part of the *IG05 - Human-Computer Interaction* course. The core objective of this project is to design and develop a graphical user interface by rigorously applying the user-centered design process studied in class. This iterative methodology encompasses four main phases: user research through targeted interviews, initial ideation via sketching, interactive prototyping, and final user evaluation.

For our project, we decided to create a game that draws heavy inspiration from the core mechanics of *Wordle*, the widely acclaimed puzzle game published by *The New York Times*.

The Original Wordle Context

To fully grasp the direction of our project, it is essential to understand the original game's premise. *Wordle* is a minimalist web-based word puzzle where players are challenged to guess a hidden five-letter word within six attempts. After each guess, the interface provides immediate visual feedback using a color-coded system:

- **Green:** The letter is correct and in the right position.
- **Yellow:** The letter is in the hidden word but in the wrong position.
- **Gray:** The letter is not in the word at all.

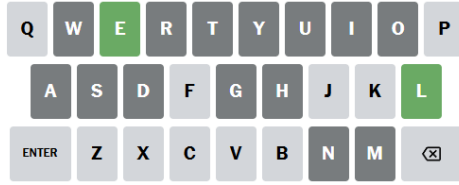
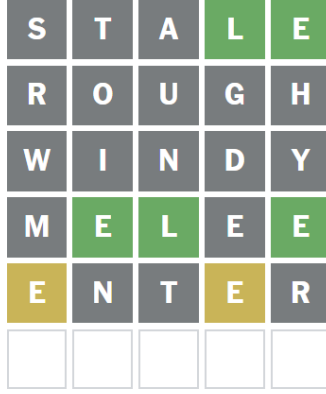


Figure 1: Game interface

A defining characteristic of the traditional *Wordle* experience is its strict daily format. Players are granted only one puzzle per day, and everyone worldwide attempts to solve the exact same word. While this "once-a-day" limitation has successfully fostered a strong sense of community and a daily ritual, it also imposes specific user experience constraints and frustrations. It is precisely these limitations that our project aims to explore, challenge, and innovate upon through the lens of human-computer interaction.

2 Users Research

At Télécom Paris, Wordle has gained significant popularity among the student body, providing us with a unique opportunity to conduct in-depth interviews. Our research focused on understanding player habits and expectations through three primary inquiries:

- Frequency of play on a weekly basis.

- Satisfaction with the current game format.
- Potential areas for improvement.

2.1 Key Findings and Insights

2.1.1 Accessibility and Engagement

The interviews revealed a dichotomy in player consistency. While many students maintain a daily streak, others find the access barrier—limited to a web browser or the New York Times application—to be a deterrent. Despite a genuine interest in the game, the lack of a more integrated or accessible mobile experience often leads to a drop in engagement.

2.1.2 Gameplay Format and Limitations

Opinions on the current "once-a-day" format are divided:

- **Minimalists:** Many players appreciate the brevity and the ritualistic nature of a single daily puzzle.
- **Enthusiasts:** A significant segment of the audience expressed a sense of being "left wanting more" (*rester sur leur faim*). These players desire longer sessions but often struggle to articulate how the gameplay could be extended without losing its core appeal.

2.1.3 The Educational Gap

A critical pain point identified by non-native speakers is the lack of context. Even when a player successfully identifies the word of the day, they are often left without its definition or usage. Currently, users must manually search external sources to understand the word they just found. This gap in the user experience became a cornerstone for our project's strategic direction.

2.2 Proposed Direction: The Rogue-like Pivot

Our research led us to explore the intersection of word puzzles and the *Rogue-like* genre. Characterized by infinite replayability, procedural generation, and an escalating difficulty curve, Rogue-likes (such as *Balatro*) allow players to accumulate bonuses and upgrades.

In this framework, the player's goal shifts from simply solving a puzzle to "breaking the game"—finding synergies between various powers to achieve

exponential scores or infinite recursive loops. This evolution addresses the demand for longer play sessions and deeper strategic engagement.

2.3 User Personas

We have synthesized these needs into four distinct personas that will guide our design process:

Persona 1: Carl (The Strategist)

A fan of Rogue-likes like *Balatro*, Carl finds the basic Wordle concept interesting but mechanically shallow. He seeks gameplay that rewards optimization and allows for "game-breaking" synergies.

- **Core Requirement:** Incorporate Rogue-like mechanics and allow for complex, "game-breaking" power-up synergies.

Persona 2: Barbara (The Language Learner)

A student using Wordle to improve her English vocabulary. She is frustrated that the game provides no definitions or semantic connections between words.

- **Core Requirement:** Integrate unobtrusive, toggleable educational tools (e.g., definitions, synonyms) to support learning without causing cognitive overload.

Persona 3: David (The Casual Player)

A fan of relaxing "clicker" games. David finds traditional Wordle stressful because it requires significant effort to recall complex vocabulary. He seeks a more streamlined, lower-friction experience.

- **Core Requirement:** Implement dynamic difficulty adjustments and assist mechanics to reduce cognitive strain while maintaining a satisfying sense of progression.

Persona 4: Anne (The Hardcore Fan)

An experienced Wordle player frustrated by the 24-hour wait time and the repetitive nature of the current format. She wants a more substantial experience that she can dive into for 15 to 60 minutes.

- **Core Requirement:** Extend the core gameplay loop and introduce diverse game modes to support longer sessions and ensure long-term replayability.

3 Design process

3.1 Initial definitions

The initial design deliberations were about elements that could give some quantitative feedback to the players, allowing them to measure their progression and improvement. For that purpose, two classical gaming elements were adopted for the project: “level progression” and “score”. At the same time, it was important to include a core element of the rogue-like genre: the “rewards”, as they keep the player engaged in the next levels, and also express their progression during the game. It was decided that a good option was to offer “power-ups”, that is, upgrades that could improve the performance of a player in a level, while also modifying the gameplay itself of the traditional Wordle. That way, the main cycle of the game would correspond to: earn points → to finish a level → to have access to more power-ups → to earn (more) points (in the next level). Another idea was to add another very common tool from rogue-like games: a “shop” to buy the power-ups, which is an interesting way to offer rewards because they give more freedom of choice to the player, while also allowing them to approach each level more strategically. Consequently, a concept of “coin” would be needed to match the idea of using a shop - and that’s why it was decided that the score, as a local (level) achievement, would reward the player with coins, which in turn would offer the player the opportunity to choose the power-ups.

3.2 Screens

With these 4 core elements established: level, score, coin and power-up, the next step was to sketch the game screens. 4 groups of sketches were made, as it is possible to see in Figures 2 to 5.

The sketches were compared at each screen proposed, so they could be compared in the same context. The main types of screen proposed were: Main Menu screen, Level screen, Results screen and Shop screen

3.2.1 Main Menu screen

Groups 2 and 4 proposed the standard menu design with the main title at the top and the menu options as central buttons, and this was kept in the final design. While the group 4 proposed to include numerical information on the front page and accessing the options as new pages, the minimalist idea of group 2 was preferable, in which less information would be displayed and the options would appear in pop-ups (blurring the background), being

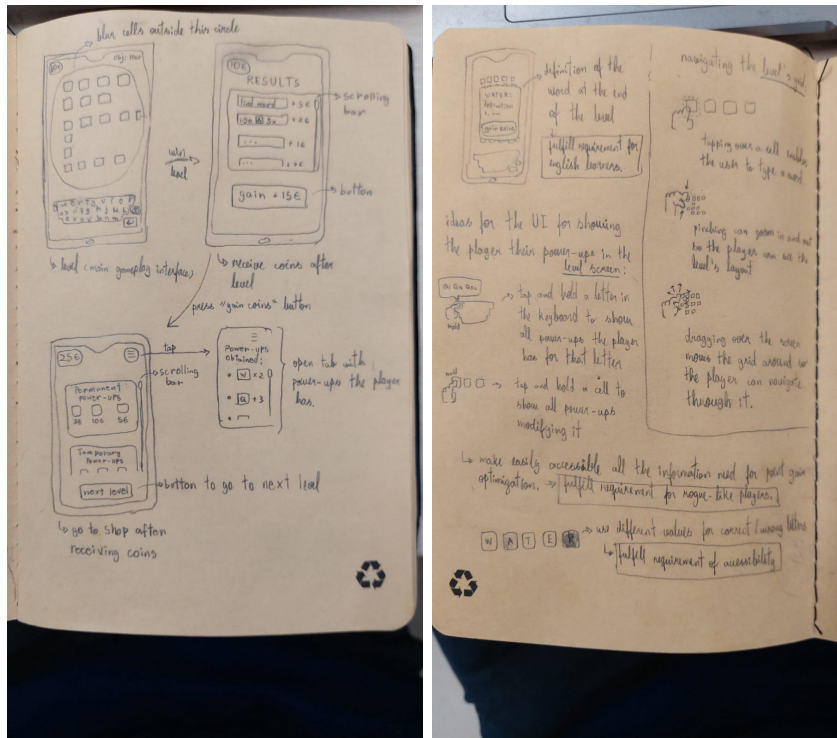


Figure 2: Group 1 of first sketches

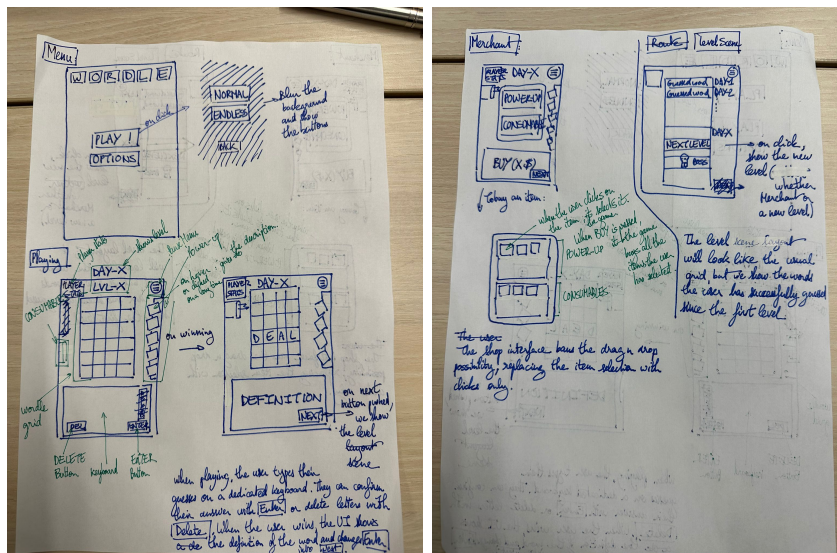


Figure 3: Group 2 of first sketches

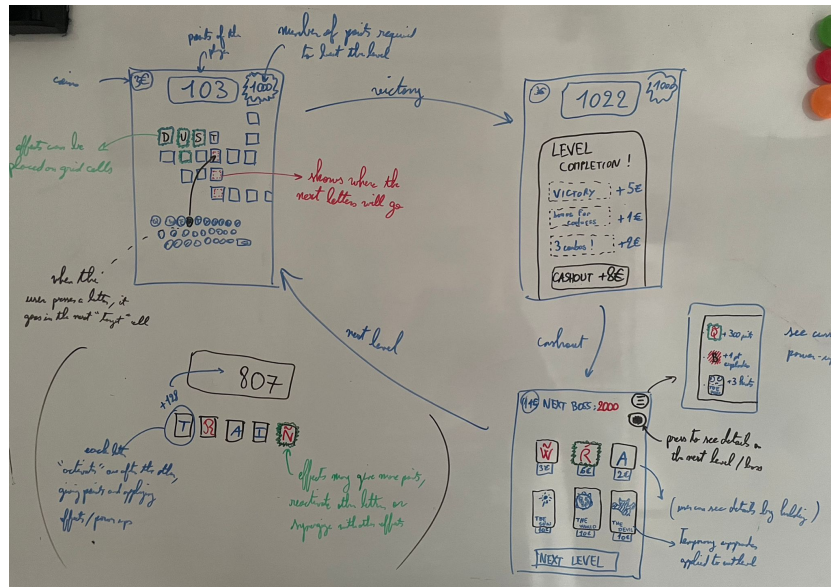


Figure 4: Group 3 of first sketches

more straightforward to the user. These pop-ups would contain a cross in the top-right to close the menu as well as a “back” button doing the exact same thing. From experience people tend to feel lost in the pause menu, and often struggle to intuitively quit the menu if the only possibility to close it is with a “resume” button, which doesn’t suggest closing as much as a red X. The best options to include in the menu were: “Continue”, enabled if there is a run saved, and allowing the player to continue this run; “New Run”, to start a new run; “Stats”, showing information such as the player’s highest score ever achieved; “Options”, with the accessibility options (explained in the next section); and “Credits”, displaying the name of the authors. The Figure 6 shows the final design idea for the Main Menu screen.

3.2.2 Level screen

All groups suggested the standard Wordle format in which the grid would appear in the middle of the screen and the keyboard at the bottom - because preserving that would make it more intuitive for Wordle players. Apart from the letters, the cells could also contain backgrounds, animations and the amount of points received by each letter would be displayed over them, drawing the attention of the player. Letters that have power-ups would have some indication of it on them, like a different font or a different shape. The keyboard should have the disposition of keys as in traditional mobile keyboards, and it was suggested that there could be an accessibility option

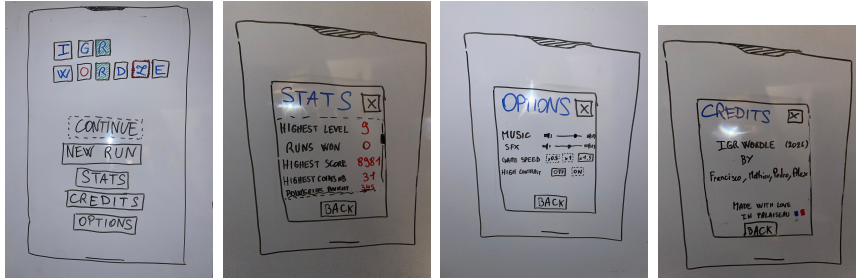


Figure 6: Final Main Menu screen

to change the keyboard between AZERTY and QWERTY (the main layouts used by the stakeholders).

Many discussions were made about the grid layout, especially about the order in which the cells could be filled. Groups 1 and 3 proposed grids that could be filled both horizontally and vertically, making the game more unique in comparison to the traditional format, while groups 2 and 4 preferred the standard rigid format. In the end, a combination of ideas was accepted: the grid would still be necessarily filled one row at the time (top-down) and one cell of the row at the time (left-right), in order to preserve the original mechanic of using the color of aligned letters to give hints about the secret word - but the amount of cells per row could variate, as long as the last row preserved the correct disposition to guarantee the player would always have enough cells to guess the correct word. Also, because the rows can be larger than the width of the screen, an option to zoom-in/out would be included. Other design decisions were made to help the player understand the layout, for example, blurring cells distant to the center, so that they do not clash with other UI elements as much, but still imply that there is information outside the screen and that the player may drag or pinch the screen to see it in focus.

At the top, each group proposed different ideas of information that could be displayed and where they could be located. The final decisions were: the score with the level number at the top center (as in group 3), being the most important information displayed during a level; pause button with access to menu options at the top left, which is a common choice in games and web pages; and the power-ups with the coin information inside an inventory tab on right side, reducing the amount of unnecessary information displayed close to the grid and still allowing the player to check their power-ups and amount of coins. In the inventory tab there would be horizontal scrollable sections in which the player would be able to see the current “build” of the

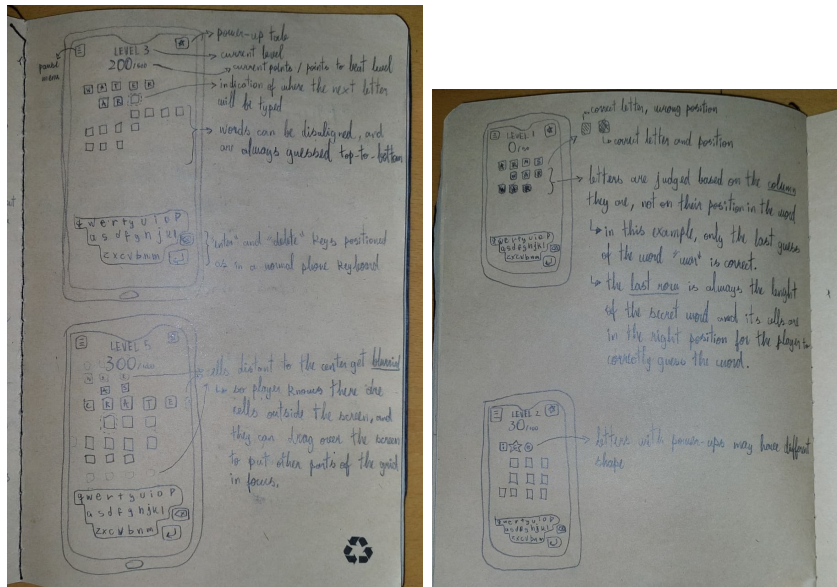


Figure 7: Final Level screen

run, which is the list of all bought (and thus active) items. The Figure 7 shows the final design idea for the Level screen.

3.2.3 Results screen

Instead of displaying many too much information as indicated on the group 4, the results screen was decided to be a pop-up combining the layouts of the group 1 of showing level statistics and also the word definition at the top: that way after each level the players would receive information about their performance on the level while also learning new vocabulary, keeping with the educational goal for the Language Learner persona. The screen would also display information (in a scrollable section) such as the challenges achieved by the player and the corresponding amount of coins. The Figure 8 shows the final design idea for the Results screen.

3.2.4 Shop screen

As proposed by all groups, the main layout of the shop should consist of a grid, in the center, of power-up options, displayed as squared icons with the prices below; the amount of coins would be displayed on the left and the “next level” button on the bottom - again, a usual layout of “rogue-like in-game shops”. Instead of separating the power-ups by category and making the screen scrollable, as in groups 1 and 4, it was decided to adopt a regular

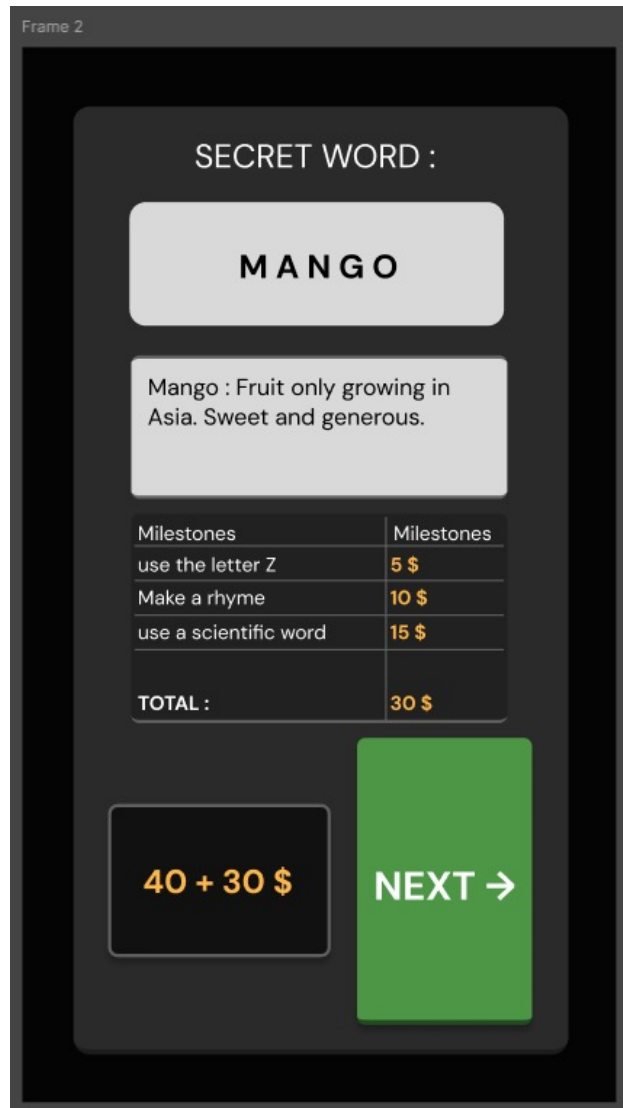


Figure 8: Final Results screen

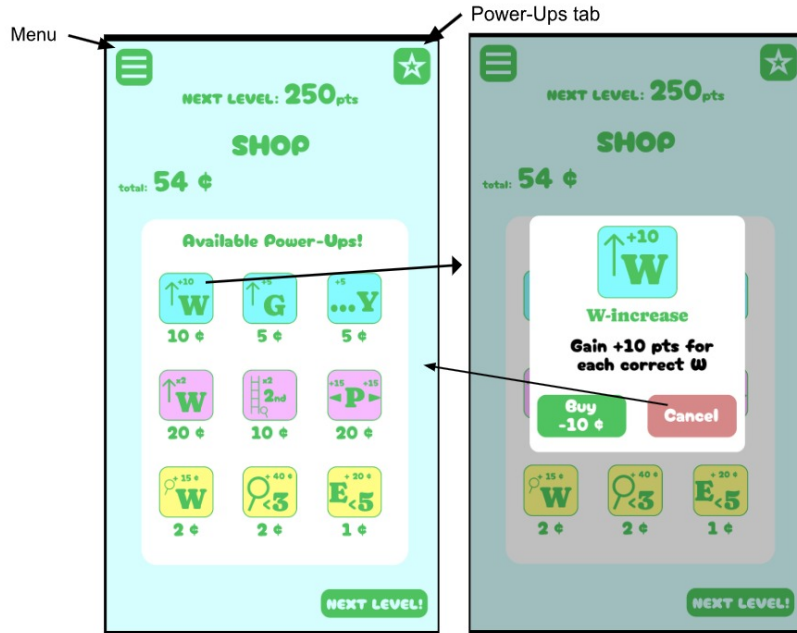


Figure 9: Final Shop screen

grid offering a small and constant amount of power-ups each time, which would variate randomly. Clicking on a power-up icon would display a pop-up showing the icon, the power-up description and the option to buy the power-up, as in group 4 sketch. The Figure 9 shows the final design idea for the Shop screen.

4 Prototype

4.1 Implemented Features

Our final prototype contains the following relevant features for our design objectives:

- An english dictionary from where we can extract words and information about them, such as their meaning.
- Infinite level generation with randomized grids.
- A screen at the end of a level (when player wins) that shows the definition of the level's secret word, even if the player couldn't guess it.

- A high-contrast option for the UI elements.
- A proper unified UI theme (including colors, fonts, styles, etc.).
- A power-up system based on combination of three types of power-ups (element, background, diacritic), and a handful of power-ups of each type with unique interactions between them.
- A shop that generates random unique power-ups by randomly combining the power-up types into a single power-up.
- VFX that give comprehensive information to the player about how the active power-ups are affecting the gameplay.
- Gesture controls to navigate the game's grid (zooming in/out, moving it around)

These features can all be considered complete on a functional standpoint, whereas in a visual aspect, some of them (mainly because of knowledge limitations regarding the framework we were using) could still be worked a bit further on.

4.2 Features we could not implement

From all the features mentioned in our document for **milestone 2**, three of them ended up not being implemented:

- The three different types of power-ups a player could buy — we only implemented one (permanent power-ups).
- A visual indication on the next cell that will be filled when the player types a letter.
- A save system for the player to stop playing at any time and then come back to where they were in a given run — it has started being implemented but is not finished yet.

All of which were important for the final product, but had less priority.

4.3 How we implemented them

The project was fully implemented using the Godot Game Engine, and as such, the user interfaces were all implemented by organizing a tree structure of visual elements in the engine's editor (similar to how QT works).

4.3.1 English dictionary

Regarding the english dictionary, it was actually taken from this GitHub repository: <https://github.com/wordset/wordset-dictionary>, which is a dictionary licensed under a Creative Commons Attribution-ShareAlike 4.0 International License. Then, to remove unwanted words (profanity), we created a python script that used a robust library to filter those unwanted words (profanity-check) and also organized each word by size so that in the game we could create a function that picked a random word of size x according to the random grid we generated. Since this dictionary also came with word definitions, we used that to implement the victory scene where the definition of the word was shown.

4.3.2 Power-ups

Then, for the power-ups, we implemented them as a combination of `enums` an object of the class `LetterBox` could have as variables. The power-ups a player has (which are all associated with a letter) are stored as a dictionary in the `RunManager` class, and every letter typed by the user in a level has its correspondent power-up. As for the effect resolution, the `LevelManager`, which is the class responsible for resolving guesses, would then apply the power-up of the letter when its position in the word was judged while resolving the guess. If the power-up resolution ended up activating other power-ups, we would then store those in a queue and keep resolving them until there were no more power-up effects to resolve, and we would pass to the judgement of the next letter in the word.

Each power-up is also linked to a unique visual representation. For each sub-category of power-up, we came up with a way to represent it in such manner that it could be combined with the other sub-categories of power-ups to create a composition of visuals that conveyed a unique power-up combination. That means that:

- For **elements**, they are represented as a texture on the letter.
- For **point modifications**, they are represent as a diacritic.
- For **area of effect**, they are represented as a background texture.



This way, not only we can provide unequivocal information about what

a power-up does, but we can also (and we actually did) link a description with each power-up. We then could create a little window explaining the power-up's effects by just concatenating the texts which describe each sub-category that composes the power-up.

4.4 Project's GitHub repository

The project's repository can be found at <https://github.com/Alex-Onceanu/IGRWordle>

5 Evaluation

We started gathering feedback on our design choices early in development. Two weeks after the project started, we already had a non-roguelike wordle working so we started asking our friends to test the game. This helped us make better choices, for instance we weren't initially displaying the remaining letters the user could still use (although the original Wordle does), which made the game harder because the user had to manually think about each letter and see if it could still be part of the mystery word.

What actually stood out is how different the feedback became when we compiled the project for mobile. In the beginning even though the project's format was vertical and made for touchscreen, there was no problem with clicking each key with the mouse, since our testers were playing the game in the Godot game engine editor. When they started testing the game on mobile, we realized for instance the buttons were way too small (we ran into the fat finger problem). On iPhone, the score was hidden by the phone's notch in the top-center part of the screen.

But testing on mobile had the advantage of making the setup easier. Instead of having to ask people if they had time to go in a separate room, to get our computer and mouse out, setup the project etc, we could now simply ask them "would you like to try the game we made ?" and hand them a phone. This led to us gathering feedback pretty much everywhere : next to the coffee dispenser, in a classroom before the teacher came, in the bus, ... Even though the rogue-like genre shines on long sessions (sessions of 15 to 30 minutes), the fact our game was separated into different "levels" implied that our friends could simply test the first four levels (which took no more than 5 minutes), and tell us what they thought of it.

Most of the feedback came after the demo in front of the class, which means we didn't yet fix every issue with our design, but this gives us good

places to start in order to improve our interface. The most common complaints were:

- The fat finger problem and score hidden by the iPhone notch
- The camera blur on the edge of the screen being too prominent
- The game not stopping even if the user had found the mystery word (for us this was initially a feature ! Since the goal is to get more points the user gets more room for combos and point farming even after the mystery word was found)
- The point system being a bit "disjoint" from the rest of the game : during the first few levels the power-ups are a bit useless because the user doesn't yet have enough for combos, which means the level is basically vanilla Wordle three times in a row, before the power-ups start giving the game its new goal.
- The rules being a bit hard to understand without having someone to explain, and the tutorial being simply text instead of an interactive experience.
- The irregular grid not being very intuitive. Some people thought they had to guess multiple words in a single line since the cells could be separated by blank spaces.

But overall people liked our ideas and enjoyed the game. After the barrier to understanding how the four elements of the power-ups work, people started getting into it and some asked if there was a possibility for infinite combos (and there actually is ! But it's a bit hard to setup (you would need a letter which reactivates another to reactivate the first) so we simply ensured this wouldn't trap the game into an infinite loop and then left this combo as it is, since our players found it fun). People especially enjoyed our non-uniform grid system, which made our game less monotone by having a different shape of level each time, and permitting 3-to-6-letter words depending on the shape of the level.

6 Conclusion

This project allowed us to explore the full user-centered design process, from initial research to prototyping and evaluation, in the context of a familiar yet constrained application: a *Wordle*-like game. By grounding our work in real user feedback, we identified key limitations of the original experi-

ence—namely its restricted accessibility, limited replayability, and lack of educational support—and used these insights to guide our design decisions.

Our main contribution lies in reimagining the traditional *Wordle* formula through the integration of *rogue-like* mechanics. By introducing systems such as levels, scores, coins, and power-ups, we transformed a short daily puzzle into a richer and more engaging gameplay loop. This shift not only addressed the needs of more dedicated players seeking longer sessions, but also opened the door to strategic depth and emergent gameplay, while still preserving the intuitive core mechanics of the original game.

Additionally, we incorporated features aimed at improving accessibility and learning, such as dictionary integration, adjustable difficulty elements, and visual feedback systems. These design choices were directly inspired by our personas and helped ensure that the experience could appeal to a broader range of users, from casual players to language learners.

The evaluation phase highlighted both the strengths and the current limitations of our prototype. While users responded positively to the originality of the concept and the depth introduced by power-ups, several usability issues—particularly on mobile—revealed the importance of iterative refinement. Challenges such as interface clarity, onboarding, and balancing the early-game experience remain areas for improvement.

In conclusion, this project demonstrates how established interaction paradigms can be meaningfully extended through user-centered design. Although the prototype is not yet fully polished, it provides a strong foundation for future development. Further work could focus on improving usability, refining game balance, completing unimplemented features, and enhancing the tutorial experience. Ultimately, our approach illustrates the potential of combining simple mechanics with layered systems to create a more engaging, flexible, and educational user experience.